

Mitigating the thermal environment in built-up local climate zones: Expanding from internal to surrounding greenspace

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ARTICLE INFO

Keywords:

Local climate zone
Urban greenspace planning
Land surface temperature
Landscape structure
XGBoost

ABSTRACT

The increase in impervious surfaces within cities has led to an increase in the urban heat island (UHI) effect, and with the shortage of available land, the effective planning of greenspace patterns is essential for developing climate-resilient cities. We adopted the perspective of local climate zones (LCZs) and utilized the ECOSTRESS land surface temperature (LST) and urban greenspace data. Based on the eXtreme Gradient Boosting (XGBoost) model and SHapley Additive exPlanations (SHAP) method, we explored the characteristics of the impact of greenspace landscape indices at two scales, namely, internal and surrounding the LCZs, on LST. We found that compact buildings (LCZ 1–3) are constrained by a lack of internal and surrounding greenspace and do not exhibit a stable dominant greenspace landscape index. In open building zones (LCZ 4–6), the dominant factors influencing LST are the Landscape Shape Index (LSI) and Percentage of Landscape (PLAND) for the internal and surrounding greenspace, respectively. The trend of the influence of LSI on LST in the two types of greenspace showed opposite effects; during the daytime, elevated LSI in greenspace internal the LCZs may lead to an increase in LST of 0.4–1.5 °C, whereas elevated LSI in the neighboring greenspace can lead to a decrease in LST of 0.4–1 °C. Furthermore, elevating the Patch Cohesion Index (COHESION) of greenspace patches helps maintain a stable cooling source. In conclusion, in this study, we reveal the mechanisms through which the internal and surrounding greenspace morphology affects LST, providing a reference for greenspace planning.

1. Introduction

Global urbanization has continued to accelerate, with the spatial expansion of cities and population agglomeration becoming prominent features [1]. According to the United Nations World Urbanization Prospects report, by 2050, 68 % of the global population will reside in urban areas [2]. High-intensity human activities lead to higher temperatures within cities, creating an urban heat island (UHI) effect. This further exacerbates the intensity and frequency of extreme heat events and poses a serious threat to human health [3–6]. Studies have shown that for every 1 °C increase in temperature, the morbidity and mortality rates for related diseases increase by approximately 1.1 % and 2.1 %, respectively; in particular, inner-city residents face higher health risks

under heat stress [7]. Urbanization has been accompanied by the replacement of large areas of natural surfaces with impervious hardened surfaces; thus, in an increasing number of people are affected by the UHI effect, which has continued to grow, with approximately 1.7 billion people globally suffering from the effects of extreme heat between 1983 and 2016 [8]. Therefore, there is an urgent need to explore and implement targeted strategies to mitigate the UHI effect and reduce its negative impacts on population health.

The UHI effect has become one of the most important climate challenges worldwide, and many scholars have investigated this issue [9–13]. This effect refers to significantly higher land surface temperature (LST) in the urban core than in the suburbs. Its core feature is significant spatial heterogeneity [14–16]. Previous studies have used the

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<https://doi.org/10.1016/j.buildenv.2025.113841>

Received 8 July 2025; Received in revised form 13 September 2025; Accepted 7 October 2025

Available online 8 October 2025

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“urban-rural dichotomy” to characterize the spatial heterogeneity of LST at the macro scale; however, this approach is inadequate for the study of fine-scale thermal environments. Therefore, some scholars have divided cities into different functional zones and analyzed the differences in LST between functional zones [17–19]. Furthermore, considering the potential influence of natural factors on UHI, some studies have explored its generalization from large-scale perspectives such as climate zones [20,21]. Urban buildings have a significant impact on temperature, and the degree of impact varies across building indicators [22,23]. Stewart and Oke proposed the classification scheme of local climate zones (LCZs) by considering the influence of urban morphology on LST. The system categorizes urban buildings into 10 types (LCZ 1–10) and land cover into seven types (LCZ A–G); each type represents a typical microclimate environment, providing an important theoretical basis and analytical framework for the study of fine-scale urban thermal environments [24].

In recent years, scholars have conducted substantial research on the urban thermal environment using the framework of LCZs [25,26]. For example, Zhang et al. showed that compact mid-rise buildings (LCZ 2) had the highest temperature during the day, whereas water bodies belonged to the lowest temperature type (LCZ G) [27]. Owing to the obvious differences in urban form and building structure between LCZs, existing studies have mostly focused on the thermal differences between different LCZs; studies have scarcely investigated the significant heterogeneity in the LST of the same type of LCZs, wherein the temperature difference within the same LCZs can exceed 10 °C, or even exceed that between different LCZs [28,29]. The temperature differences within the LCZs category mainly stem from two aspects: first, the thermal characteristics of the building materials themselves, such as dark colored building materials that heat up more by absorbing a large amount of solar radiation [30]. Second, they are influenced by the surrounding environment, such as buildings near parks or water bodies, which generally exhibit lower temperatures, whereas buildings near industrial areas have relatively higher temperatures [31,32].

Vegetation is a nature-based cooling method that can reduce the surrounding temperature through shading and transpiration. Numerous studies have investigated the cooling effect of vegetation, and scholars have quantified the cooling ability of vegetation from different perspectives, such as through remote sensing spectral indices, landscape patterns, morphological patterns, and vegetation heights [33–37]. However, current methods to improve the cooling effect of greenspace are based on the urban scale, and according to the principle of division, greenspace within localized climatic zones are usually less in number; the cooling effect of greenspace is also affected by different building types, and the effect of greenspace on LST in this context is unclear. Considering that large greenspace usually have long cooling distances [38], studies on planning internal greenspace and effectively utilizing large surrounding greenspace to mitigate the localized thermal environment are lacking. This gap hinders our understanding of mitigating localized thermal environments and limits the search for targeted strategies for their mitigation. To fill this gap, in this study attempt to address two research questions:

- (1) What are the characteristics of the landscape indices of internal and surrounding greenspace with respect to localized LST?
- (2) How do greenspace landscape indices affect LST differently during the day and at night?

To address the above research questions, we consider Shanghai as an example and use the city's LCZ maps and ECOSTRESS LST data to analyze the differences in LST within and between LCZ classes, and elucidate the influence of internal and surrounding greenspace on LST based on the XGBoost-SHAP framework. Our results will provide a basis for the planning of Shanghai and other cities.

2. Study area and datasets

2.1. Study area

Shanghai (120°52'–122°12' E, 30°40' to 31°53' N) is located in East China, covering an area of approximately 6340.5 km². Shanghai is located in the subtropical monsoon climate zone, which is characterized by high temperatures, rainy summers and cold winters with little rain. The average annual temperature is 18.8 °C, whereas the yearly precipitation reaches 1280.6 mm (<http://sh.cma.gov.cn/>). Shanghai is a typical representative of mega cities in China and abroad. It has undergone rapid urbanization, which is a process that many Asian and global southern cities are currently experiencing. In 2023, the urbanization rate of Shanghai was 89.46 % (<https://jrj.sh.gov.cn/index.html>). Therefore, the urban thermal environment problems and their driving mechanisms discovered in Shanghai have important reference significance for other large cities in similar development stages. Additionally, the hierarchical spatial structure of “central city-suburbs-distant suburbs” formed in Shanghai is a classic model for the development of many single center cities; thus, the research results are valuable for reference in structurally similar cities. The study area is the main urban area of Shanghai, which is located within the fourth ring road and has an area of approximately 2955.5 km² (Fig. 1).

2.2. Datasets

The datasets used in this study were Landsat 8–9, ECOSTRESS LST products, and urban greenspace data (Table 1). Landsat 8–9 is a Landsat satellite data processing product provided by the United States Geological Survey (USGS). It belongs to level 2 data in collection 2 (<https://earthexplorer.usgs.gov/>), which is radiometrically, geometrically, and atmospherically corrected, and can be directly used for surface parameter inversion it was used for the classification of LCZs in this study. ECOSTRESS is a high-resolution thermal infrared radiometer deployed by the National Aeronautics and Space Administration (NASA) at International Space Station (ISS) in 2018, with a time-access period of 3–5 days. The LST product in ECO2LSTE (<https://lpdaac.usgs.gov/>), with its high spatial resolution and ability to observe multiple transits in a single day, provides key data support for fine-scale analysis of the daily characteristics of surface temperature. To reduce the impact of weather conditions and ensure data accuracy, we screen the images based on the criteria of clear and cloudless skies or low cloud cover. Simultaneously, the image transit time must be different time points, such that we can analyze the changes in LST at different times. Considering the availability of data, we selected four time points to analyze daily changes in LST in Shanghai City (Table 1). As the image at 12:09 is in October, to quantify the impact of seasonal factors on temperature deviation, we used Moderate Resolution Imaging Spectroradiometer (MODIS) data to calculate the difference in LST between October 3 and August in Shanghai. The results showed a coefficient of determination R² of 0.79 and root mean square error (RMSE) of 0.43, indicating that the deviation in LST between October 3 and August is small and can support the comparison between results (Fig. S1). Urban greenspace data were mapped using a deep learning framework based on Google Earth imagery and city boundary data with a spatial resolution of approximately 1.1 m, providing urban greenspace data at a granular scale with an overall accuracy of 87.56 % [39]. In this study, we analyzed the effect of greenspace on LST within an LCZs.

3. Methodology

Fig. 2 illustrates the technical flowchart of this study. We first collected Landsat 8–9 remote sensing images, ECOSTRESS data, and urban greenspace data of Shanghai. Subsequently, the data were pre-processed, including the classification of LCZs based on remote sensing images, geo-localization and calculation of LST data, and calculation of

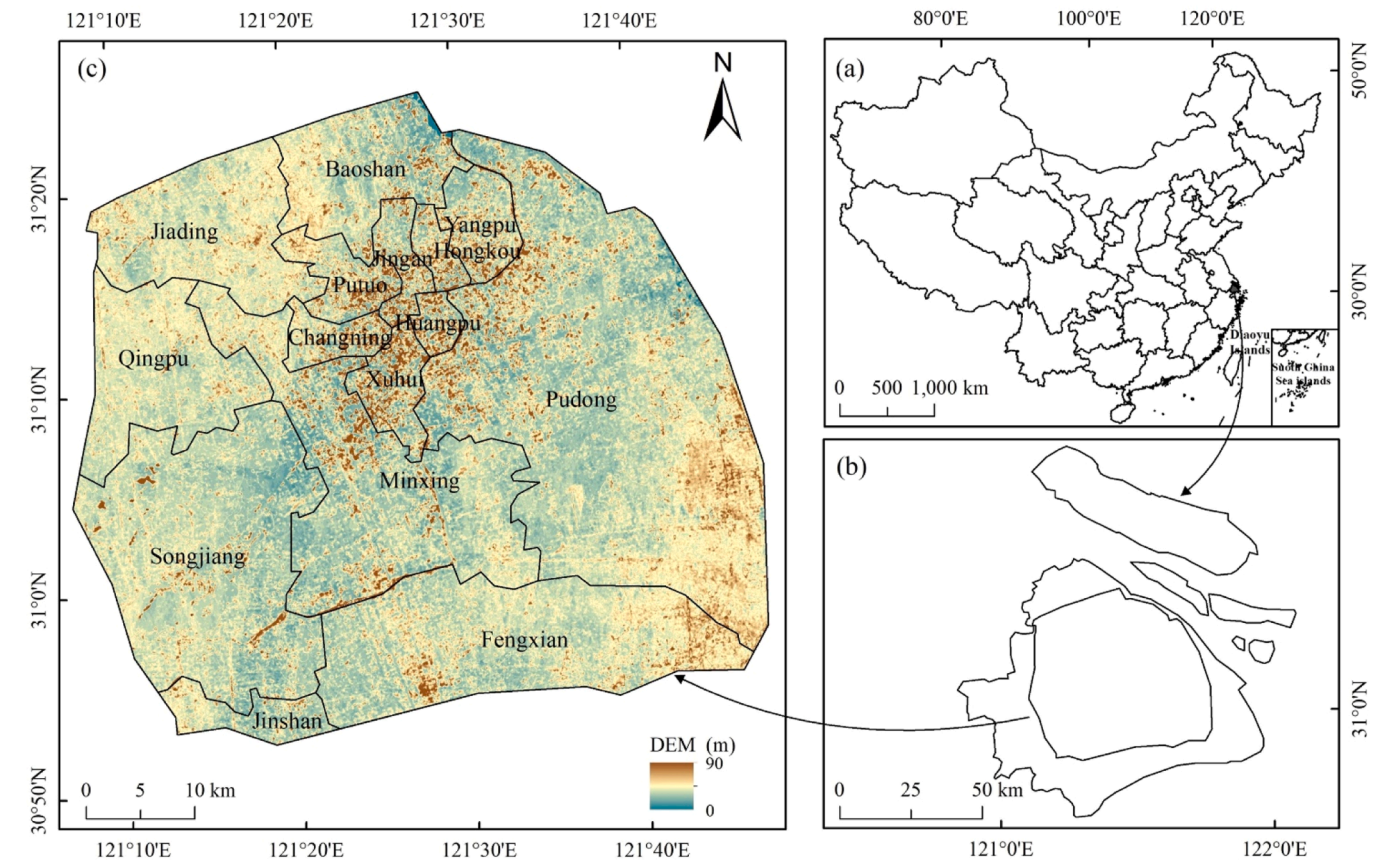


Fig. 1. Location of the study area. (a) Location of Shanghai in China. (b) Location of the research area in Shanghai and its administrative regions. (c) Scope and elevation of the research area.

Table 1
Details of the used datasets.

Data	Date	Local time	Resolution(m)	Sources link
Landsat 8-9 ECO2LSTE and ECO1BGEO	2022.08.14		30 m	https://earthexplorer.usgs.gov
	2021.08.27	02:39	70 m	https://lpdaac.usgs.gov
	2024.08.16	05:09		
	2021.10.03	12:09		
	2024.08.02	19:56		
Urban Greenspace	2020		1.1 m	https://www.scidb.cn/en/detail?dataSetId=36af2aed281e4c82aa8a3cd3f1211a37

landscape indices of urban greenspace. High-resolution urban greenspace data were used to characterize the greenspace within the LCZs, whereas categories LCZ A–D were used to describe the greenspace surrounding the LCZs. In the data analysis stage, the spatial distribution pattern and occupancy of LCZs are first examined to evaluate the heterogeneity of LST within and between LCZs classes. To quantify the effect of greenspace landscape indices on the LST of internal and surrounding LCZs and reveal the nonlinear mechanism of its action, the XGBoost model was used in this study and interpreted in conjunction with the SHAP method.

3.1. Classification of LCZs

LCZs spatially categorize regions exhibiting consistent urban fabric attributes, geometric configurations, and human-induced thermal behaviors. Stewart and Oke classified local climate zones into 17 types. There are three commonly used methods for delineating LCZs: remote sensing, geographic information system (GIS), or a combination of the two. In this study, a remote sensing-based method was used for LCZs

delineation. First, the training samples are divided in Google Earth; the number of samples of each LCZ is maintained at not less than 50 to ensure classification accuracy. LCZ 7 was not considered in this study owing to its low number in Shanghai. Subsequently, bands 1–7 and band 10 in the contemporaneous remote sensing image were resampled to 100 m, which is consistent with the resolution of the default LCZs maps; moreover, the cloud coverage of the remote sensing image was less than 5 %, which meets the quality requirements of classification. Finally, the resampled bands and divided samples were classified in SAGA-GIS using a random forest model, which is often used in LCZs classification and has shown good accuracy and robustness. As the unequal quantity of each LCZ type was different, after the classification, we generated a certain percentage of random sampling points based on the number of each LCZ to evaluate the classification accuracy of LCZs.

3.2. Selection of the greenspace landscape index

Landscape patterns refer to the spatial structural characteristics presented by the spatial combination, distribution, shape, size, and

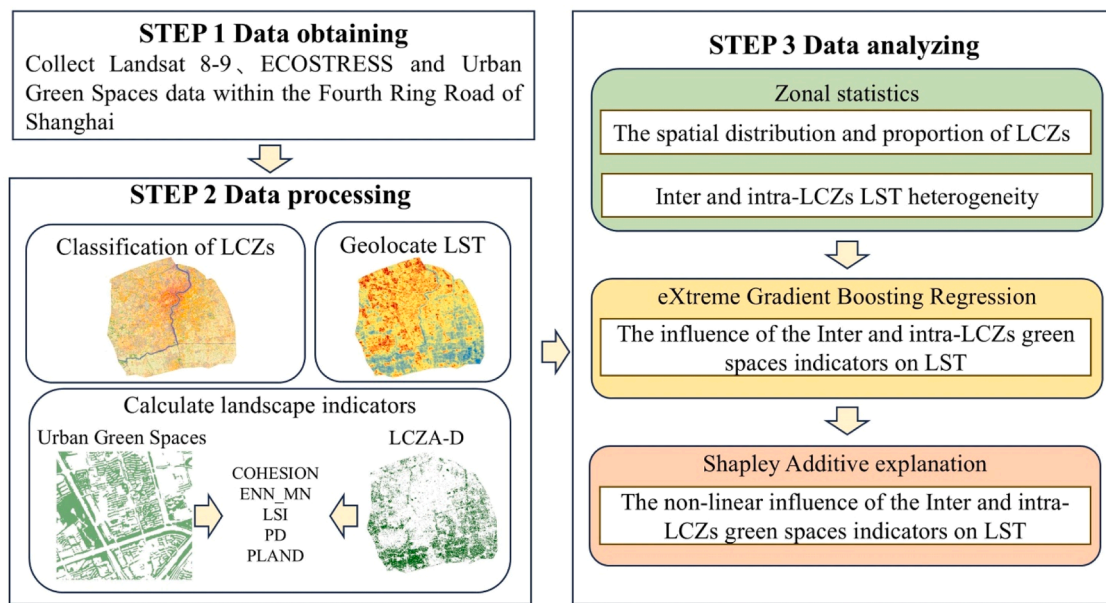


Fig. 2. Study flowchart.

Table 2
Detailed information on selected indicators.

Category	Indicators	Range	Description
Area	PLAND	[0, 100]	Refers to the area of a target patch type as a percentage of the overall landscape area.
Shape complexity	LSI	[1, max]	Used to measure complexity of plaque shape.
Connectivity	ENN_MN	[0, max]	Refers the average of the distances between all patches in a given patch type and their nearest neighboring patches.
Aggregation	COHESION	[0, 100]	Used to characterize the clustering of the distribution of individuals within the same type of patch.
Fragmentation	PD	[0, max]	Refers to the number of patches contained within a unit of landscape area.

interrelationships of landscape elements in a certain region. They reflect the spatial heterogeneity and spatial configuration patterns formed under the joint action of natural processes and human activities [40]. Within the context of increasing UHI effect and decreasing available land, landscape patterns are widely used to plan greenspace to mitigate thermal environmental problems. Greenspace indicators were selected in this study based on the following principles. First, the selected indicators have been widely used in the study of UHI problems. Second, the theory of indicators can be applied to actual planning. Third, the indicators can help characterize the greenspace distribution from multiple dimensions without redundancy. In our study, an array of greenspace landscape indices were broadly categorized as area, shape complexity, connectivity, aggregation, and fragmentation. The greenspace area has been widely recognized as being directly related to the cooling effect [41]. Shape complexity enables for heat exchange and thus LST regulation through edge areas [42]. Highly connected greenspace can create ventilation corridors that promote cool air flow and therefore result in LST relief [43]. Highly aggregated greenspace have a more stable ability to maintain low LST [44], and fragmented greenspace are susceptible to the surrounding thermal environment and their cooling capacity may be diminished [45]. Finally, the metrics chosen for this study were the Percentage of Landscape (PLAND), Landscape Shape Index (LSI), Mean Euclidean Nearest Neighbor Distance (ENN_MN), Patch Cohesion Index (COHESION), and Patch Density (PD), which were calculated using Fragstats 4.2. Detailed descriptions of these metrics are presented in Table 2.

3.3. Statistical analysis

Traditional regression models cannot easily describe the complex link between the greenspace index and LST, whereas machine learning

as a nonlinear model can satisfactorily capture the relationship between them; therefore, it has been widely used in recent years to study the drivers of UHI. To analyze the effects of internal and surrounding greenspace indices on LST, we used the XGBoost [46] and SHAP [47] frameworks. XGBoost has become one of the preferred algorithms for structured data modeling owing to its efficient parallel computation, strong regularization against overfitting, and flexible tree structure optimization. Considering the possibility of multicollinearity among landscape indicators, to ensure the reliability of the results of the regression model, we used collinearity diagnosis in SPSS 26 to analyze the relationship between various landscape indicators. The variance inflation factor (VIF) of these indicators is less than 5 and the tolerance is greater than 0.1, indicating the absence of multicollinearity (Table S1 and Table S2). Subsequently, the XGBoost model in Python is used to train the internal and surrounding greenspace metrics to analyze the properties that affect LST. With respect to the influence of surrounding greenspace metrics, we considered LCZ A-D to be the same greenspace and only considered the influence of greenspace within 300 m around building type LCZ 1–6, and excluded the data if a water body was present within this range.

SHAP is an approach derived from the Shapley value theory of cooperative game theory that aims to provide explanations for the predictive results of machine learning models and quantify the individualized contribution of each feature to the predictive results. This helps in understanding the particular roles of each characteristic in model predictions, particularly for complex black-box models [48]. In this study, the dependence_plot function built into SHAP was used to calculate the nonlinear adjustment of greenspace indicators of LST.

4. Results

4.1. LCZs map

Through the random forest model, we classified the four rings of Shanghai into 16 LCZs types; subsequently, we used 817 random points to assess the classification accuracy through the confusion matrix. The Kappa coefficient is 0.85, indicating high classification accuracy.

The spatial differentiation pattern and area share of LCZs in the central city of Shanghai, respectively, are shown in Fig. 3 and presented in Table 3. The results show that architectural types account for 76.4 %, whereas natural-like types constitute 23.6 %. Compact buildings accounted for 7.1 % of the total, with their primary distribution observed in the central area. The open building type LCZ 5 had the largest area share of 20.4 % and was primarily distributed in the western and eastern edge areas of the compact building type. LCZ 4 had an area share of 16.1 % and was more distributed in the eastern area of the compact building type, whereas the spatial distribution of LCZ 6 extended primarily to the natural-like type. The largest area in the natural-like type was LCZ D (13.2 %), which was primarily distributed in the south. LCZ B was mainly distributed in the east (5.4 %), and both LCZ C and LCZ F had a very small area share of less than 1 %.

4.2. Heterogeneity of LST between LCZs

Fig. 4 shows the LST variations of LCZs at four specific times, as well as the LST differences between LCZs. Fig. 5 shows the deviation between LST of each LCZ and the average LST of the entire city. The redder the color, the higher is the LST of LCZs compared with the city average, and the bluer the color, the lower is the LST of LCZs compared with the city average.

As shown in Fig. 4, the LST of each LCZ exhibited a significant daily variation. The daily variation of architectural LCZs was generally greater than that of natural-like types, with LCZ 8 showing the largest daily variation of 13.55 °C, and the daily changes of LST of compact buildings and industrial zones reached more than 11 °C. The smallest daily variation of 9.18 °C was observed in LCZ 6 of the open type buildings. Natural-like types LCZ E and LCZ F had the highest daily

variation of more than 10 °C, and LCZ G had the smallest at 6.09 °C.

During the daytime, the LST of the built types was usually higher than that of the natural-like types. The maximum differences in daytime LST between the LCZs, namely, 7.25 and 3.04 °C, were observed at 12:09 and 19:56, respectively. The smallest differences in nighttime LST were observed at 2:39, namely, 0.42 °C. At 12:09 and 19:56, the LST of LCZ 8 was the highest, those of LCZ E and LCZ F were the highest, and that of LCZ G was the lowest. At 5:09, LCZ 1 had the highest temperature, followed by LCZ 2, which also had a relatively high LST; at 2:39, LCZ G had the highest temperature.

As shown in Fig. 5, the deviation between LCZs and the urban average LST is minimal at 02:39, approaching the urban average LST. Some building types have negative deviations (LCZ 1, LCZ 2, LCZ 4, and LCZ 10), indicating that the average LST of these types is lower than the urban average at 02:39. The deviation of LCZ G is 0.23 °C, which is higher than the average LST of the city. The deviation between LCZs and the average LST of the city is most significant at 12:09, with LCZ 8 showing a significantly higher average LST than that of the city, followed by compact buildings and LCZ 10. Open buildings have a similar average LST to that of the city. The deviation of LCZ G is −3.63 °C, which is significantly lower than the average LST in the city.

4.3. Impact of greenspace indicators of internal and surrounding LCZs on LST

To further analyze the influence of greenspace indicators of internal and surrounding LCZs on LST and identify the most important factors affecting temperature, we trained the XGBoost regression model at four time periods. The R^2 and RMSE of all regression models are presented in Table S3 and S4. Although the data volume of a single LCZ is relatively small compared with the sample size of the entire city, the R^2 of each LCZ is above 0.5, and the RMSE fluctuation range is less than 1 °C, providing a high level of explanatory power for the analysis results of each model. Figs. 6 and 7 show the ranking of the importance of the greenspace of internal and surrounding LCZs on the LST, respectively, with the Shapley values of the features and influencing factors on the horizontal and vertical axes, respectively. Within LCZ 4–6, the Shapley value of LSI was higher than those of other factors at different times

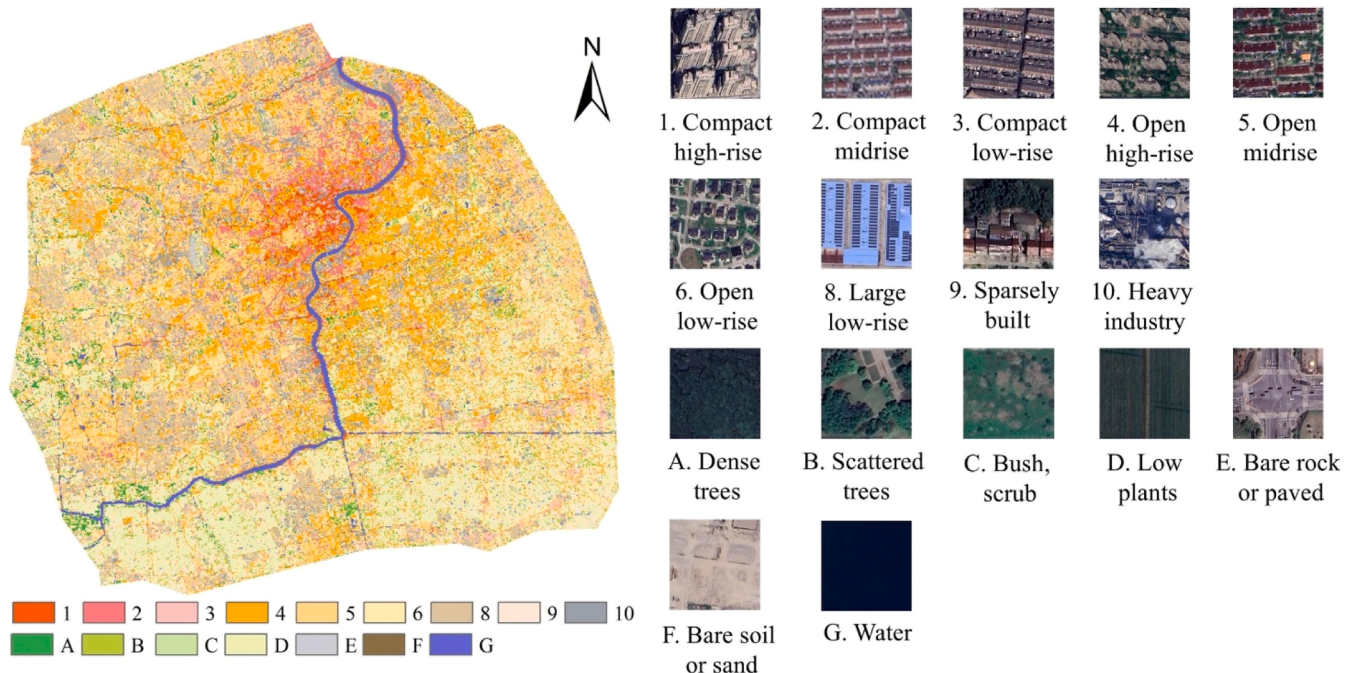


Fig. 3. Spatial distribution and definition of the LCZs.

Table 3
Area share of LCZs.

Built types	LCZ 1	LCZ 2	LCZ 3	LCZ 4	LCZ 5	LCZ 6	LCZ 8	LCZ 9	LCZ 10
76.4 %	1.6 %	3.4 %	2.1 %	16.1 %	20.4 %	11.0 %	7.6 %	6.4 %	7.8 %
Natural-like types	LCZ A	LCZ B	LCZ C	LCZ D	LCZ E	LCZ F	LCZ G		
23.6 %	1.3 %	5.4 %	0.6 %	13.2 %	1.0 %	0.0 %	2.1 %		

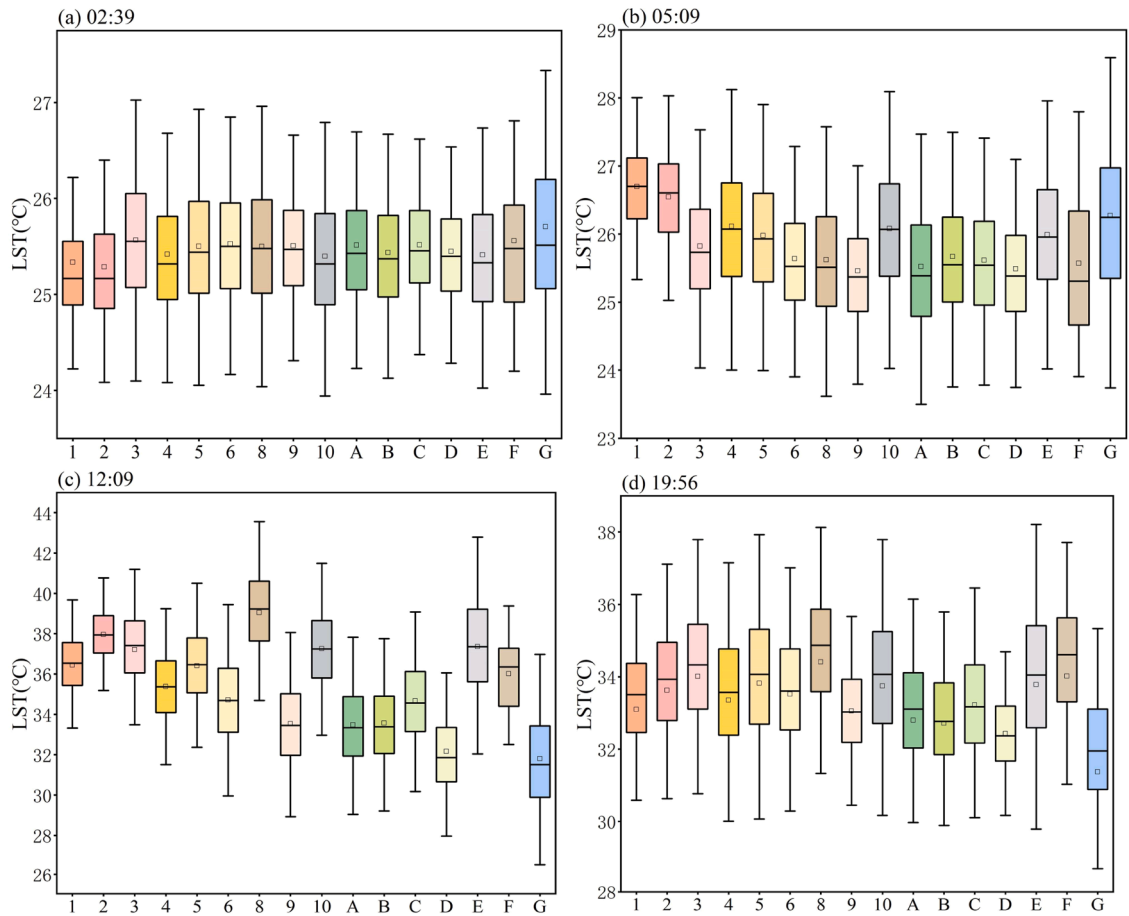


Fig. 4. Daily variation of mean LST in different LCZs. Note: The local sunrise and sunset times were approximately 5:14 and 18:45, respectively.

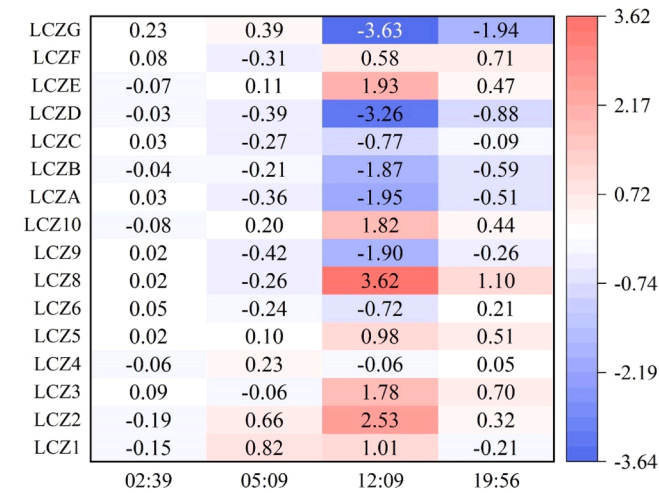


Fig. 5. Deviation between LCZs LST and urban average LST.

(Fig. 6). LSI was the most important feature and most stable factor influencing the LST of LCZ 4–6. However, in LCZ 1–3, the most important indicators affecting LST changed at different times. For example, within LCZ 1, the most important indicators at 2:39, 5:09, and 12:09 are ENN_MN, PLAND, and COHESION, respectively; and in LCZ 2, the most important indicator at 5:09 and 12:09 is PLAND. Moreover, PLAND has a critical influence on LST; it was also the most important indicator besides LSI, and the index was often ranked in the top three.

Subsequently, similarities were observed between the effects of the surrounding and internal greenspaces on LST (Fig. 7). For open buildings, the indices that affected LST the most were relatively consistent, whereas for compact buildings, variations were observed across the LCZs and at different times for the same LCZs. In particular, for open buildings, PLAND was the most important and consistent index; whereas for compact buildings, PLAND was the most important index at 5:09 and 12:09 for LCZ 1, and only at 12:09 for LCZ 3. Furthermore, the most important factors affecting the LST of LCZ 2 were PD and LSI, respectively. ENN_MN of the internal and surrounding greenspace was the least influential factor for temperature, often located ranked in the last place. Furthermore, instability was observed in the influence of PD and COHESION on temperature, with frequent and significant changes in the

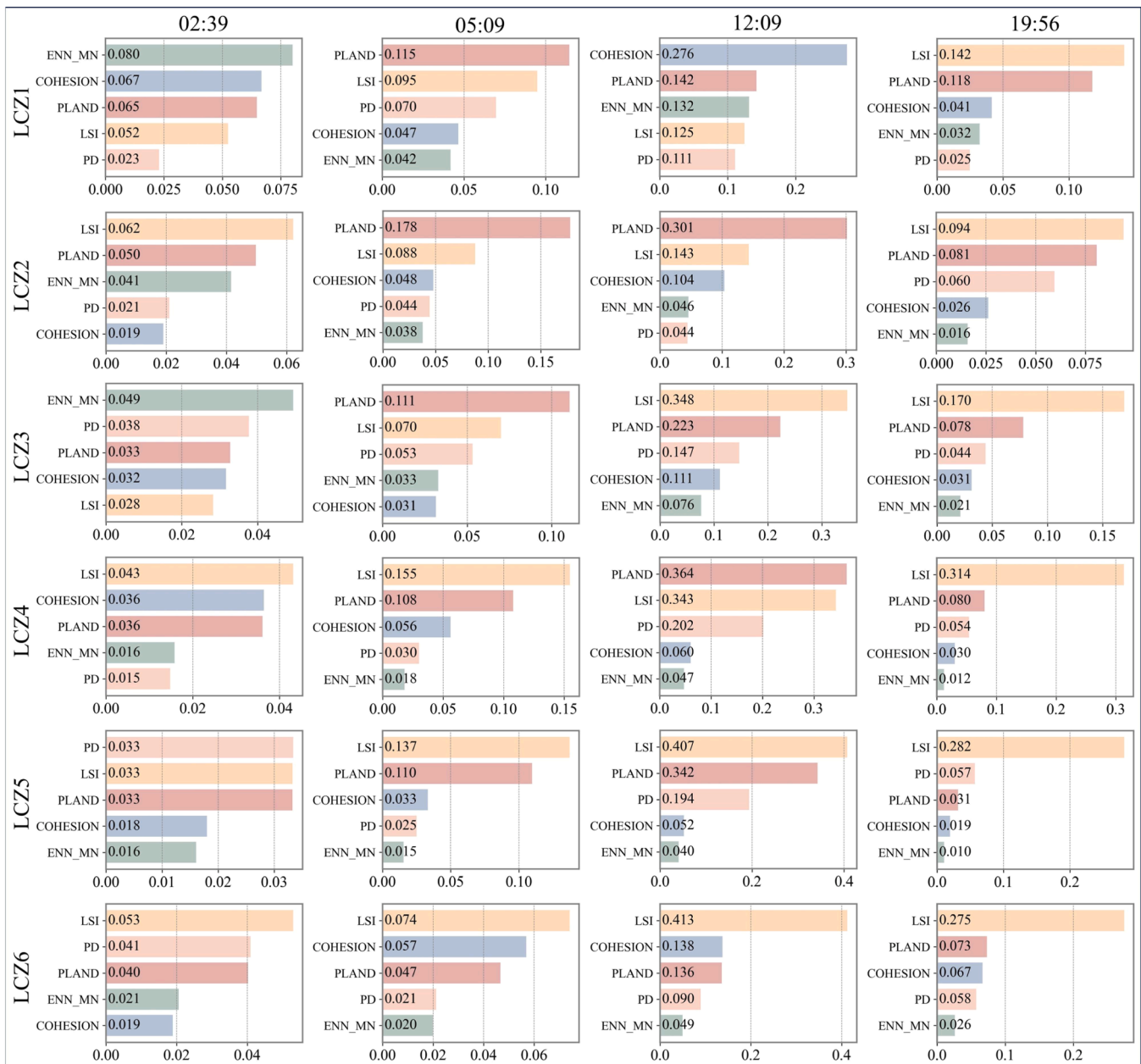


Fig. 6. Ranking of the importance of greenspace internal LCZs for LST at different times. The values on each bar chart represent the Shapley values of the indicators. The larger the value, the greater the impact of that indicator on the LST.

order of importance of the two indices at the four time periods.

4.4. Nonlinear effects of greenspace indicators on LST

To understand the nonlinear effects of greenspaces of internal and surrounding LCZs on local temperature, we selected 12:09 and 2:39, respectively, to represent the impact of greenspace on day and night LST. Fig. 8(a) and (b) show the nonlinear effects of internal and surrounding greenspace at night, and Fig. 9(a) and (b) show the nonlinear effects of internal and surrounding greenspace during the day, respectively; the horizontal and vertical coordinates of the scatter plots represent the ranges of each metric and degree of the affected LST, respectively.

The trends in the effects of the internal and surrounding greenspace metrics on the LST of different LCZs were not always consistent; for example, PD exhibited both warming and cooling effects. Additionally, indicators varied in the extent of their influence, showing a greater

range of influence on LST during the day than during the night. For instance, in PLAND, the magnitude of the internal greenspace regulating LST during the day was, on average, 0.74 °C higher than that at night, compared with it being 1.7 °C higher in the surrounding greenspace.

The effect of ENN_MN on LST was extremely weak in both the internal and surrounding greenspace, with no clear trend of increasing or decreasing LST with increasing ENN_MN. COHESION generally had a cooling effect on LST, ranging from 0.04 to 0.4 °C. During daytime, the PLAND of the two greenspace generally had a cooling effect, but a significant difference was observed between the cooling effect of PLAND of the interior and surrounding greenspace. Moreover, according to the PLAND of the surrounding greenspace during daytime, the cooling range was 0.92 °C higher than that of the interior greenspace, whereas during nighttime, the PLAND of the two greenspace had a warming effect for some LCZs. The trend of the effect of LSI on diurnal LST was the same for both greenspace but the internal and surrounding greenspace showed the opposite trend. For example, during daytime, the LSI of the internal

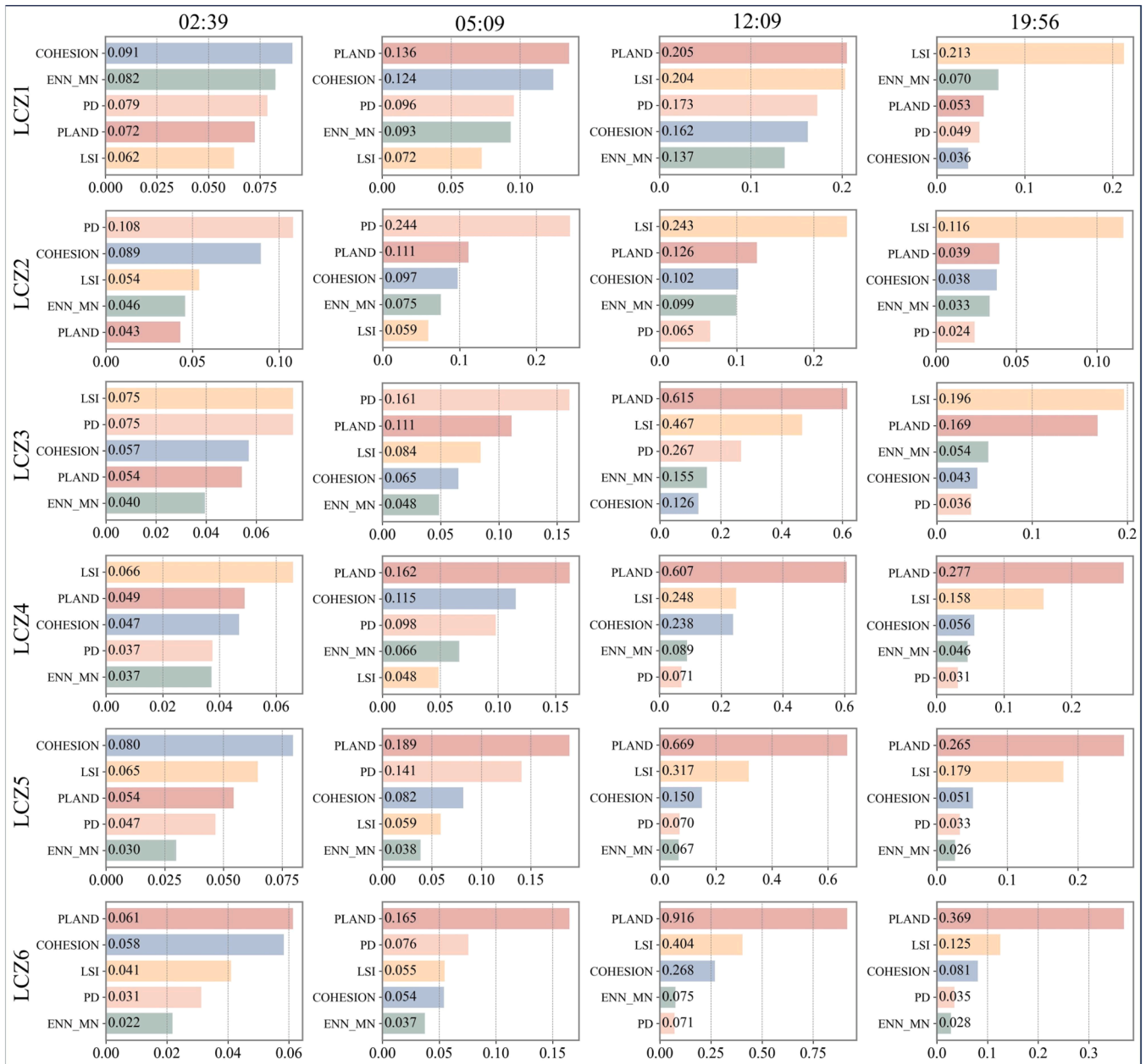


Fig. 7. Ranking of the importance of LCZ surrounding greenspace for LST at different times. The values on each bar chart represent the Shapley values of the indicators. The larger the value, the greater the impact of that indicator on the LST.

greenspace had a warming effect with a warming range of 0.4–1.5 °C, whereas that of the surrounding greenspace had a cooling effect with a range of 0.4–1 °C.

5. Discussion

5.1. Understanding LST heterogeneity between LCZs

The LCZs framework, which divides cities into 17 unique climate zones based on similarities in surface properties, indicates that significant LST differences exist between different types of zones. During the day, as buildings absorb more heat compared with vegetation and water bodies, the LST of built-type LCZs is typically higher than that of the natural-like type [49,50]. However, at night, the LST of natural-like type LCZs can be higher than that of some building types. As shown in Fig. 4 (a) and (c), water exhibits a strong cooling effect during the day owing to its high specific heat capacity; conversely, at night, its slow heat release

often causes its temperature to be higher than that of the rapidly cooling surrounding built-up areas, turning it into a relatively warm area, which is consistent with the results of comparison between day and night in the LCZs of Fuzhou [51]. This phenomenon highlights the dynamic complexity of LST research and emphasizes the limitations of analyzing at a single time point [52]. By focusing on high-LST LCZs types to propose targeted mitigation strategies, we found that LCZ 8 had the highest LST during daytime in this study, which is consistent with the findings of Lin et al. [53]. The thermal storage capacity of LCZ 1 in compact buildings during the day (under solar radiation) are lower than those of LCZ 2 and LCZ 3, making LCZ 2 and LCZ 3 hotspots in cities. Among natural-like types, LCZ A-D has lower LST owing to vegetation transpiration and shading, whereas LCZ E and LCZ F mostly contain bare surfaces with less vegetation, resulting in higher LST, which is consistent with the results of Dalian [54]. In comparison, the characteristics of thermal environments of different LCZs in Shanghai have certain similarities with those of other cities. Therefore, using Shanghai as the

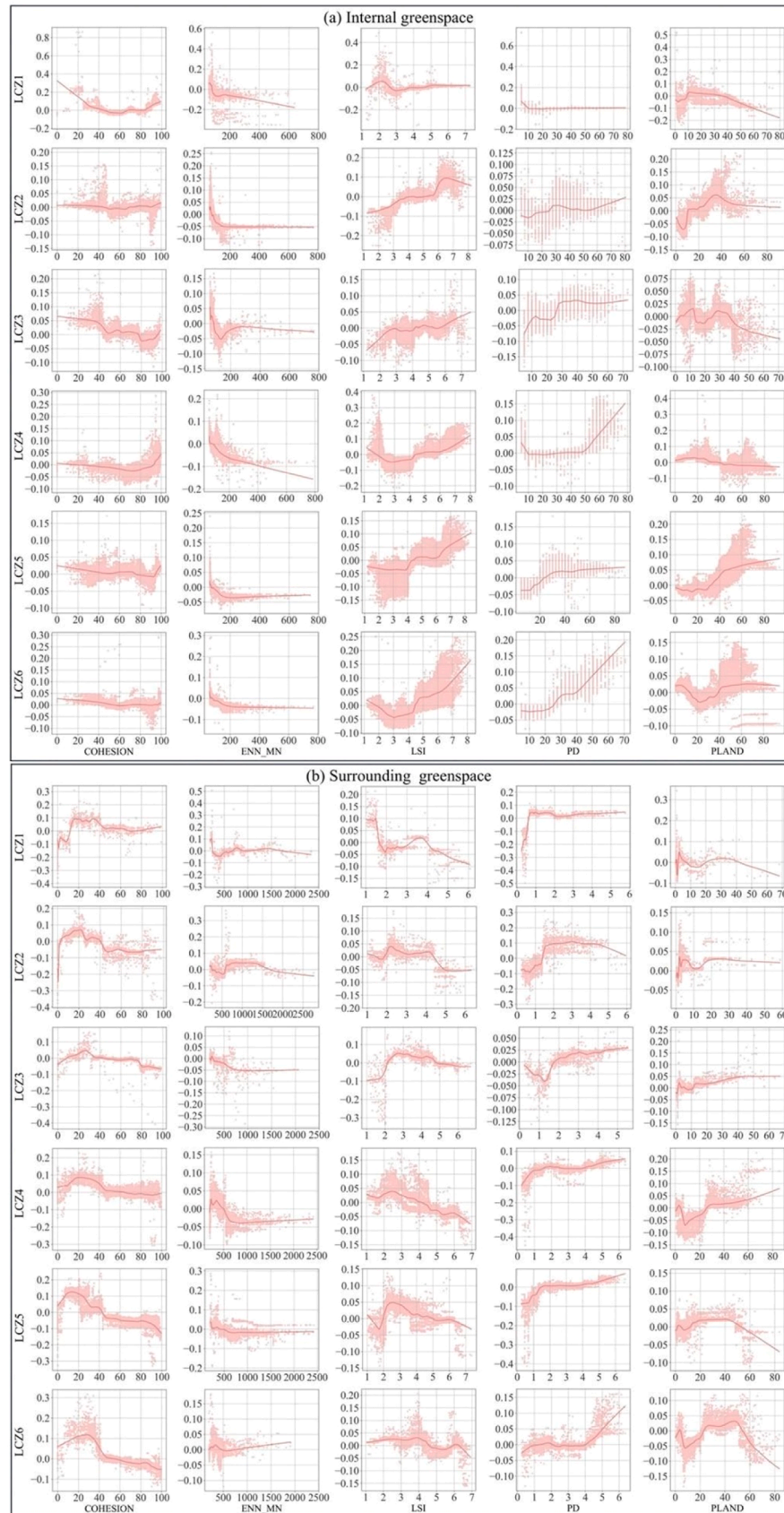


Fig. 8. Nonlinear relationship between greenspace and nighttime LST. (a) and (b) are internal and surrounding greenspace, respectively.

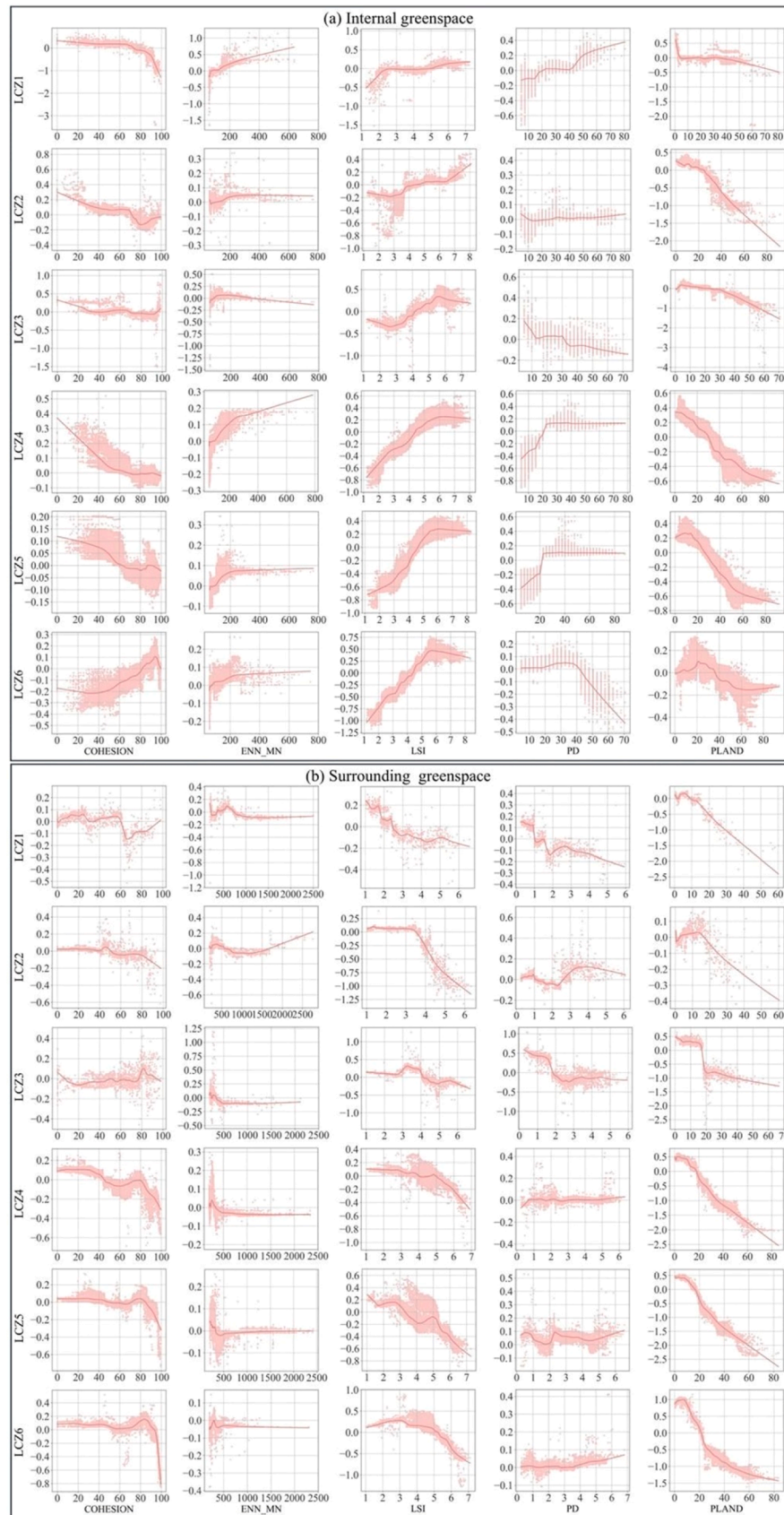


Fig. 9. Nonlinear relationship between greenspace and daytime LST. (a) and (b) are internal and surrounding greenspace, respectively.

research area has universality and can provide certain reference value for the planning of other cities. By calculating the deviation between the average LST of LCZs and the city, we can understand the contribution of each LCZ to UHI. As shown in Fig. 5, the deviation of each LCZ is very small at night (02:39), approaching the average LST of the city. However, during the day (12:09), the deviation of LCZs from the average LST of the city increases. Compact buildings, LCZ 8, and LCZ 10 are the main contributors to UHI, whereas the LST of open buildings are similar to the average LST of the city. LCZ A-D and LCZ G are cold island areas in the city. Therefore, in urban renewal, the planning of open buildings and blue-green space should be prioritized to create a comfortable living environment.

5.2. Characteristics of internal and surrounding greenspace indices affecting daytime and nighttime LST

To mitigate the local thermal environment, we aimed to find greenspace planning strategies that are favorable for cooling in both the internal and surrounding LCZs. A direct link exists between the greenspace area and cooling effect [55–57]. However, inside LCZs, the area of greenspace patches is usually limited. Hence, the greenspace patches make the LSI a core element influencing LST through a complex edge effect, which is especially prominent in open buildings. This implies that the shape of greenspace has an obvious moderating effect on LST through the planning of greenspace of internal open buildings. However, in compact buildings, the dominant elements influencing LST at different points in time are different, mainly because the greenspace in compact buildings is small and easily disturbed by the surrounding environment, hindering the formation of stable influencing factors. The influence of greenspace on LST in surrounding LCZs has a similar characteristic, and there are no fixed influencing factor around compact buildings; however, around the open type buildings, the PLAND of the greenspace is the most important indicator influencing the temperature. This is because when the greenspace is larger, the influence of the PLAND on LST exceeds that of the other configuration factors of greenspace [58].

The importance of characteristics enables us to understand the extent to which each metric of the greenspace affects LST from a general perspective. However, it does not enable us to understand the direction in which it affects the temperature. In contrast, Figs. 8 and Fig. 9 show the nonlinear influence of LST in LCZs by various factors. Most previous studies have shown that PLAND in greenspace can reduce LST [59,60]. The same conclusion was obtained in this study; however, in this study, the PLAND of the two types of greenspaces appeared to increase LST at night (Fig. 8(a) and (b)). This may be because greenspace with high PLAND values are adjacent to large areas of hardened surfaces, which have a reduced cooling effect at night, and heat absorbed by the hardened surfaces during the daytime is released at night, which results in an increase in LST with increasing in PLAND [61]. COHESION may have a cooling effect, which is consistent with previous results [62,63]. In this study, there was bidirectional modulation of the effects of PD on LST; the role of PD has been controversial in previous studies, with some suggesting that PD can have a cooling impact [64,65], whereas others suggest the contrary [66]. A higher PD implies that the number of greenspace patches is higher. In contrast, the edge density of the patches increases, and the contact area between the patches and surrounding environment increases. At this time, one of the two situations may be present: (1) the contact area increases, thus absorbing more heat from the surroundings to achieve the cooling effect; and (2) if the area of the greenspace patches is too small, the internal microclimate regulation ability is weak and will instead be interfered with by the surrounding thermal environment, resulting in warming. The increase of LSI will similarly lead to increases in the contact and surrounding areas. The area of the internal greenspace is small and easily affected by the surrounding hot air. Hence, the LSI of the internal greenspace of LCZs had a warming effect (Figs. 8(a) and Fig. 9(a)), whereas the surrounding

greenspace has a larger area, and the increase of LSI could instead further reduce the surrounding LST (Figs. 8(b) and Fig. 9(b)). This finding highlights the importance of the internal and surrounding greenspace, and that of differentiated planning.

5.3. Mitigation strategies

Based on our study, we have presented several targeted mitigation strategies. As the LST of compact buildings (LCZ 1–3) are usually higher than those of open buildings (LCZ 4–6), in the process of urban renewal in Shanghai, old houses or old residential areas can be transformed into open buildings rather than into compact buildings. This is beneficial for urban ventilation and reducing heat accumulation. The LST of the same LCZs can also be highly different. Therefore, from the perspective of the building itself, the use of highly reflective building materials or construction of green roofs can be encouraged to prevent the building from absorbing large amounts of heat. From the perspective of external factors, the thermal environment can be mitigated by greenspace, and, from an internal perspective, urban greenspace planning can be initiated from the internal and surrounding greenspace of LCZs, to improve the aggregation degree of greenspace patches; a high degree of aggregation of patches can serve as a stable cooling source, and maintaining the shape of greenspace patches can decrease the interference of surrounding heat. Owing to the relatively small greenspace area inside LCZs, especially in compact buildings, the thermal environment of LCZs of building types can be mitigated by surrounding greenspace regions, such as building parks, which can mitigate the thermal environment by constructing a network of cold islands. These greenspace regions usually have a larger area, which increases the cooling capacity of the surroundings by improving their complex shapes.

5.4. Limitations

Based on the LCZs framework, we analyze the impact of greenspace on local LST at both internal and surrounding scales. Although we provide specific mitigation strategies, there are certain limitations that need to be addressed. First, owing to the availability of data, LST data were derived from ECOSTRESS data on different dates, which affected the accurate determination of the daily variation in LST in our analysis. Future research should focus on analyzing the deviation between LST on different dates to enhance the robustness of analysis and credibility of the conclusions. Second, LCZ A–D was not studied separately in this study owing to limitations in sample size, and some studies have shown differences in cooling effects of different vegetation types; therefore, the study area could be expanded in future studies to meet a sufficient sample size to analyze the cooling effects of each natural-like type. Third, the synergies between the size of greenspace patches and other configuration indicators are unclear, namely, how to plan the shape of patches of different sizes to minimize interference from the surrounding environment, maintain a stable source of cooling, or improve the cooling effect on the surroundings. Finally, we consider Shanghai as a case study and preliminarily reveal the impact mechanism of internal and surrounding greenspace on the LST of LCZs. However, the formation of the urban thermal environment is the result of multiple factors. In future studies, systematic sensitivity analysis can be conducted in cities of different climate zones and scales to further quantify the impact of two types of greenspaces on the LST of LCZs under different urban backgrounds, and obtain more universal conclusions.

6. Conclusions

In this study, the daily variation of summer LST within and between LCZs was investigated based on the LCZs framework with Shanghai as the study area. The characteristics of the landscape pattern of internal and surrounding greenspace affecting the LST were analyzed, and the main conclusions are summarized as follows:

- (1) The built-up areas in the study area accounted for 76.4 % of the total area, whereas the natural-like areas only accounted for 23.6 %. This presented a layered distribution structure from the inside out, transitioning from compact and open built-up areas to natural-like areas, reflecting the highly urbanized land use characteristics.
- (2) The diurnal variations in LST of built-up areas (especially LCZ 8, compact built-up areas, and industrial areas) are significantly higher than those of natural-like areas. Among them, the daytime warming effect of LCZ 8 is the most prominent. The LST differentiation among LCZ types is most obvious at 12:09 but significantly weakens at 2:39 when the temperature of LCZ G is the highest. This indicates that the heterogeneity of the thermal response of the underlying urban surface is closely related to solar radiation and the redistribution of surface energy.
- (3) The greenspace coverage within and surrounding compact buildings (LCZ 1–3) is relatively low, hindering the formation of a stable influencing factor. In contrast, the LSI of greenspace within open buildings (LCZ 4–6) plays a dominant regulatory role in their thermal environment, whereas the PLAND of surrounding greenspace becomes a key factor influencing the thermal effect of this type of area. ENN_MN had no significant overall effect on LST.
- (4) Greenspace have a greater range of LST regulation during the day than at night; however, differences in the trend of the effect can occur. COHESION can have a cooling effect during both the day and night. ENN_MN did not show a significant warming or cooling trend, whereas PD showed both warming and cooling effects. In general, PLAND has a warming effect during the night and a cooling effect during the day, and based on PLAND in the surrounding greenspace, the cooling range is 0.92 °C higher than that of the internal greenspace. The LSI of the internal greenspace of LCZs can warm up by 0.4–1.5 °C during the daytime, whereas that of the surrounding greenspace can cool down by 0.4–1 °C. This indicates that the spatial configuration of greenspace had an environmentally dependent and complex impact on the UHI effect.

In conclusion, in our study, we investigated the properties of the greenspace of internal and surrounding LCZs to regulate LST. Moreover, we suggest targeted strategies to mitigate the localized thermal environment problems. Future studies should be extended to different cities to verify the generalizability of the results and investigate the cooling effects of different vegetation types to provide a more comprehensive approach to climate adaptation strategies.

Funding sources

This research study was supported by Liaoning Revitalization Talents Program (grant no. XLYC2202024), Basic Scientific Research Project (Key Project) of the Education Department of Liaoning Province (grant no. LJKZ0964), Fundamental Research Funds for the Central Universities (grant no. N2111003), Natural Science Foundation of Guizhou Province (grant no. (2019)1150), High-End Cultivation Program of Liaoning Normal University (grant no. 24GDL004).

CRedit authorship contribution statement

Xuefeng Kang: Writing – original draft, Methodology, Formal analysis. **Jun Yang:** Supervision, Project administration, Funding acquisition. **Yuqing Zhang:** Software, Resources, Methodology. **Zhe Li:** Visualization, Validation, Resources. **Yanchu Feng:** Methodology, Investigation, Formal analysis. **Xiangming Xiao:** Validation, Supervision, Methodology. **Jianhong (Cecilia) Xia:** Visualization, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to acknowledge all colleagues and friends who have voluntarily reviewed the translation of the survey and the manuscript of this study. We would like to thank Editage (www.editage.cn) for English language editing.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.buildenv.2025.113841](https://doi.org/10.1016/j.buildenv.2025.113841).

Data availability

Data will be made available on request.

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